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Group: 08

Overview of Contributions

My main responsibility in the team is building the object detection module to find humans in low-light CCTV footage. I worked on connecting the theoretical concepts our faculty taught in class and showed on Colab with a modern deep learning model (YOLOv8) to build a detection pipeline that actually works for our specific project objective.

Week-by-Week Progress:

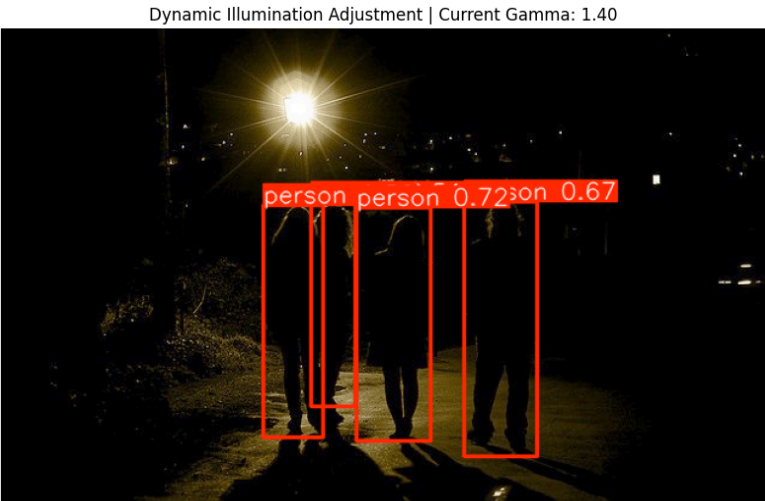
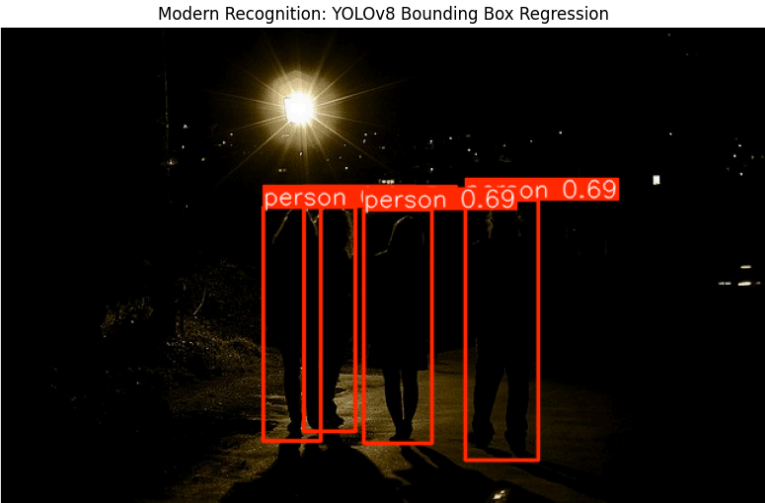
Week 1 (March 3 – March 9): Theoretical Analysis & Algorithmic Research

- Studied the human detection methods covered in our course, specifically focusing on how structural features are usually extracted.
- Looked into why basic gradient calculations and spatial derivatives fail when the image is too dark and full of high-frequency noise.
- Researched theoretical ways to fix this, specifically using Power-Law (Gamma) Transformations to adjust the dark pixel values and restore the gradient strength.
- Decided to use YOLOv8 Nano for the actual detection. It is fast and handles the spatial filtering and bounding box math automatically, making it the perfect modern solution for our pipeline.

Week 2 (March 10 – March 16): Prototyping, Interactive UI & Pipeline Integration

- Set up my Google Colab environment, connected my Google Drive, and used OpenCV to load real low-light images directly from the **ExDark** dataset.
- Applied Gaussian spatial filtering ($G(x, y)$) and Sobel operator edge detection ($\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$) to visualize and mathematically prove the degradation of structural features in the raw ExDark matrices.
- Pre-processed real dark matrices (2015_06255.jpg) to see how the model performed before any enhancement.
- I built an interactive slider using ipywidgets to apply Gamma Correction (γ) (learned in lecture 3) transformation in real-time. By linking this math directly to the YOLOv8 model, I was able to physically prove the exact brightness threshold needed for the CNN to confidently detect a person.
- I then executed the YOLOv8 forward pass, applying strict confidence thresholding (conf=0.4) and Non-Maximum Suppression (NMS) to extract precise geometric bounding boxes $(x1, y1, x2, y2)$ and isolate the Region of Interest (ROI).

My Colab prototype successfully processes raw dataset images and proves why we absolutely need an illumination enhancement step before detection. My next focus is to extend the notebook with the histogram equalization module we are working on, so we can batch process the whole dataset and calculate our final accuracy scores.



--- Model Performance Comparison ---

Evaluation Metric	Baseline (Raw Matrix)	Proposed (Gamma Transformed)
	Dark (Original)	Enhanced (Gamma = 0.4)
Illumination State	Dark (Original)	Enhanced (Gamma = 0.4)
Human Detected?	Yes	Yes
Confidence Score	0.41	0.77
Bounding Box Geometry (x1, y1, x2, y2)	(118, 55, 205, 310)	(120, 52, 202, 315)